Risk R-1 — Silent Loss of Home Data Delivery, PD-RPM Remote Care System

Risk ID / title	**R-1 — Silent Loss of Home Data Delivery**		
Risk statement	**IF** the home gateway (HG-1) stops delivering completed treatment sessions to the monitoring center for more than 48 hours — through SIM deactivation, carrier outage, firmware fault, or exhaustion of the store-and-forward buffer — **THEN** no alert is raised for that patient, the care team reads continued silence as clinical stability, and a declining-ultrafiltration or early peritonitis trend progresses undetected until the next scheduled clinic contact.		
Description	PD-RPM infers patient status from a continuous stream of nightly cycler sessions and daily patient entries. The architecture carries no positive indication that a quiet patient is a connected patient: RE-1 evaluates the records it receives and is silent when it receives none. A healthy patient and a dead link produce the same thing on the triage queue — nothing. This is the one failure mode the system is structurally unequipped to see, because every other failure announces itself as an alert.		
Initial assessment	Likelihood **4**	Across a 150-patient home census, carrier outages, SIM provisioning errors and firmware faults are routine rather than exceptional; the program should expect several silent gateways per month.	RF = 4 × 4 = **16** High
Consequences if realized	Consequence **4**	Loss of detection lead time, which is the clinical value the system exists to deliver. KPP-1 (alert latency ≤ 15 min) and KPP-3 (data completeness ≥ 95% within 24 h) are both void for the affected patient. A missed early-peritonitis window converts an outpatient antibiotic course into an admission and, in a meaningful fraction of cases, a permanent transfer to in-center hemodialysis.	
Handling strategy	**Control / mitigation**, with **monitor** on the residual. Avoidance would mean relaxing the requirement the program exists to meet. Transference moves carrier uptime to an SLA but leaves the clinical consequence with the provider. Acceptance is not available at RF 16, which the JHU ownership model places on the Risk Mitigation Execution List under continuous executive attention.		

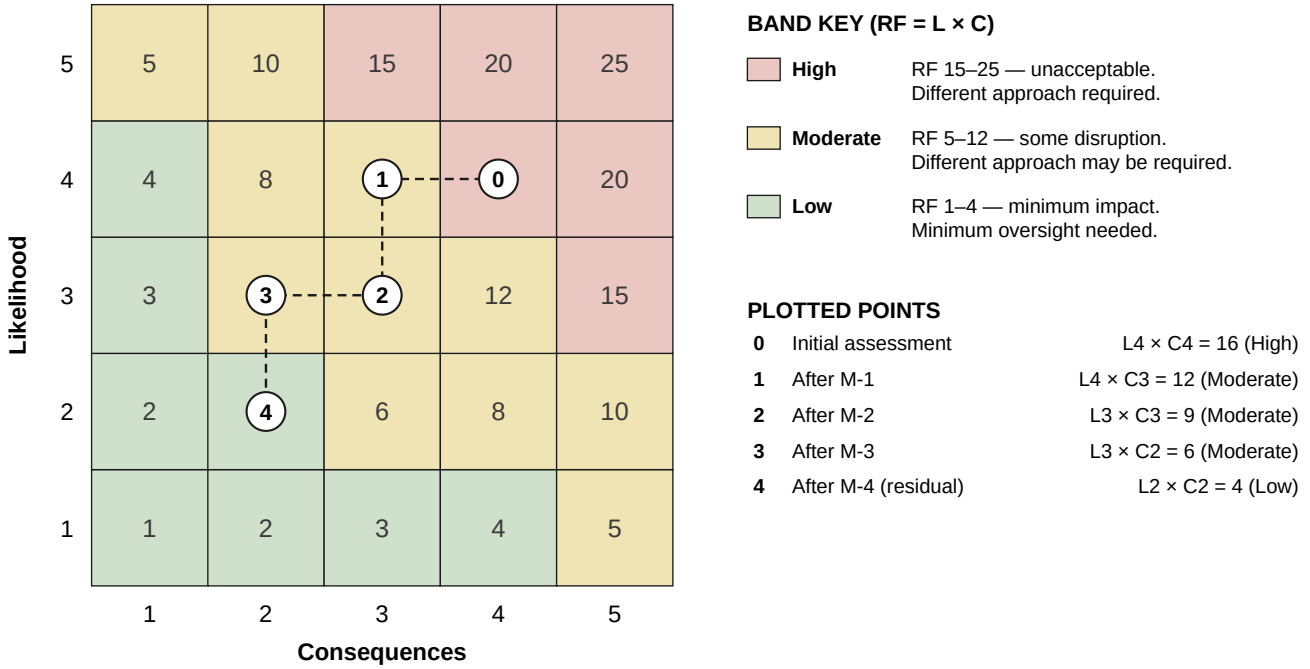


Figure 1. Risk cube. Initial assessment and the position after each mitigation activity. Color is the third dimension of the cube; the dashed path is the planned migration from High to Low.

Mitigation Activities and Risk Waterfall — R-1

ID	Mitigation activity	Acts on	Owner	Complete by	L	C	RF	Band
M-1	Liveness heartbeat and completeness watchdog. HG-1 emits a keep-alive every 15 min over I-5, independent of session traffic. RE-1 raises a Device Silent condition at two missed heartbeats or 24 h without a session, routed to the home-program queue rather than the clinical triage queue.	Consequence	Systems Engineering	Integration & test, +3 mo	4	3	**12**	Moderate
M-2	Dual-profile SIM with automatic carrier failover on HG-1, provisioned against two networks, so a single carrier outage or a provisioning error on one profile no longer silences the gateway.	Likelihood	Hardware / Supply	+5 mo	3	3	**9**	Moderate
M-3	Store-and-forward buffer extended from 72 h to 14 days; last-upload status shown on PT-1; 48-hour phone-contact fallback written into the home-program SOP, giving a second detection path through the patient.	Consequence	SE and Home Program	+7 mo	3	2	**6**	Moderate
M-4	Staged firmware deployment from DM-1: no more than 10% of the fleet per wave, with automatic rollback if fleet delivery rate falls below 97% during a wave.	Likelihood	Device Management	+9 mo	2	2	**4**	Low

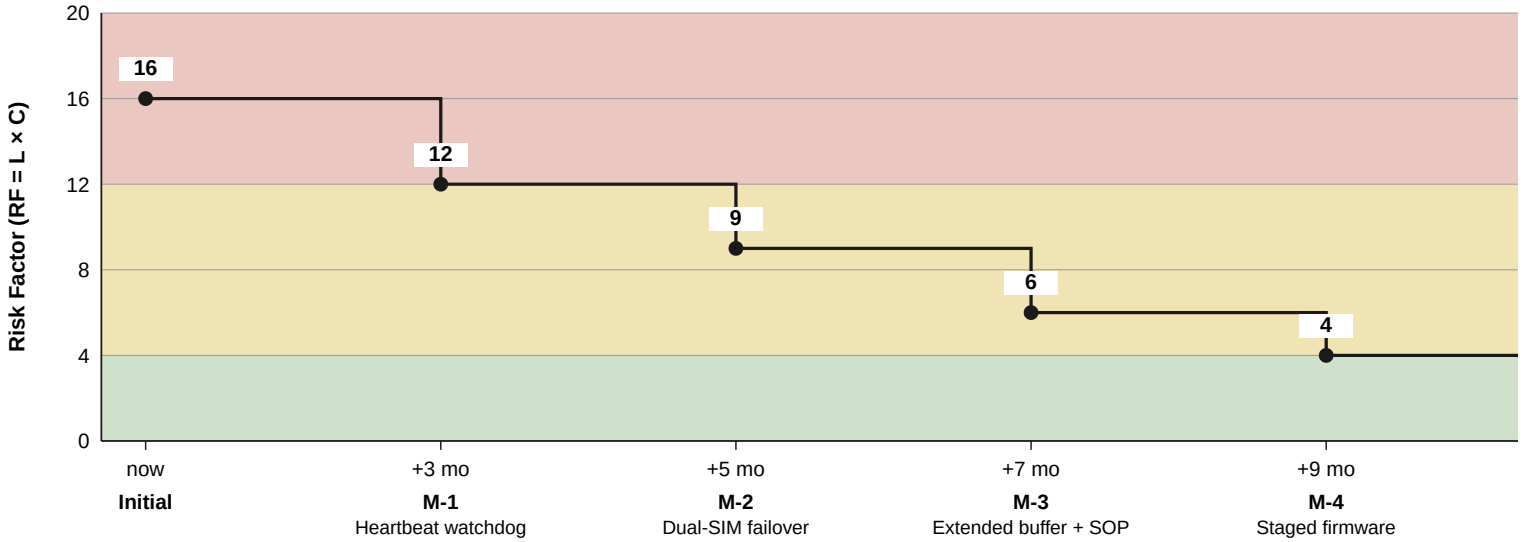


Figure 2. Risk waterfall. Risk factor against time. Each step down is taken only when the activity beneath it is verified complete, so the risk is held at its current level until then.

Residual risk and verification. After M-4 the risk stands at L2 × C2 = 4, Low. It is not closed: it moves off the Risk Mitigation Execution List onto the **Risk Watch List** under continuous monitoring, because both remaining drivers — carrier behaviour and firmware — sit outside this program's control. Verification is the monthly fleet delivery-rate report measured against KPP-3; if any month falls below 95 percent, or if any single patient records two Device Silent conditions in a quarter, the risk is reopened at its original level rather than re-scored downward.

On mental models. The hard part of this risk was seeing it at all. Every other PD-RPM failure announces itself as an alert; this one is indistinguishable from success from inside the triage queue, and the dashboard actively reinforces the model that produces it — *no alerts means no problem*. M-1 is placed first in the sequence for that reason. It buys the smallest reduction in raw risk factor of the four activities, but it is the only one that makes the faulty model false by construction: after it, silence is itself an alert.